# SafeMind

Product Requirements Document (PRD) — MVP 1.0

**Product:** SafeMind | **AI Safety & Intelligence Layer:** SPAG

**Initial Market:** University of Lagos (UNILAG) | **Type:** AI-assisted mental wellness and human-support platform

**Version:** 1.0

## 1. Product Vision

SafeMind is a privacy-first digital platform creating a trusted pathway between emotional distress and appropriate support. It provides a private place to express feelings, access wellness resources, receive non-clinical AI support, and connect with peers or qualified professionals when needed.

**Core principle:** AI scales support. Humans provide care.

## 2. Problem Statement

Students may face academic pressure, examinations, financial hardship, family or relationship problems, isolation, burnout, anxiety and other forms of emotional distress. Even where counselling exists, stigma, limited awareness and access friction can prevent early help-seeking.

SafeMind reduces that friction and creates an earlier, safer route to support.

## 3. Target Users & Stakeholders

- **Students (18+):** primary users seeking emotional support or connection to help.
- **Peer supporters:** trained listeners providing non-clinical support under safety policies.
- **Licensed professionals:** counsellors, psychologists or other approved providers.
- **University administrators:** aggregate operational insights only; no unrestricted access to private conversations.

## 4. Product Goals

- Create a trusted and private environment for students to seek support.
- Reduce friction between emotional distress and appropriate help.
- Provide immediate, non-clinical AI-assisted support.
- Identify interactions that may require human attention using risk/triage signals.
- Enable structured escalation to peer and professional support.
- Build privacy and cybersecurity into the product from the beginning.
- Measure whether users reach appropriate support and complete relevant next steps.
- Validate the product through a controlled UNILAG pilot before wider expansion.

## 5. Non-Goals & Safety Boundaries

- No diagnosis of depression, anxiety or other mental-health conditions.
- No claim of certainty in predicting suicide or other crises.
- No replacement of psychologists, psychiatrists, doctors, emergency services or university health services.
- No autonomous emergency contact unless an explicitly approved policy, legal basis and mechanism exists.
- No sale of identifiable mental-health data.

- No unrestricted administrator access to private user content.

## 6. Core User Journey

**Student → Secure Platform → AI Support → Risk/Triage → Appropriate Support → Follow-up**

Every interaction should move the user toward appropriate support rather than deeper isolation.

## 7. MVP Feature Requirements

| Feature | Requirement |
|---|---|
| Authentication | Secure registration/login, sessions, logout, recovery and optional MFA. |
| Privacy Mode | Anonymous/pseudonymous participation where appropriate, with clear consent and privacy controls. |
| Mood Check-in | Simple mood entries, timestamps, trends and user-controlled history. |
| Private Journal | Create, view and delete personal entries with strong access controls. |
| SafeMind AI Assistant | Empathetic, non-clinical support with clear boundaries and escalation behavior. |
| SPAG Intelligence Layer | Analyze permitted signals for safety classification, distress indicators and triage routing. |
| Human-in-the-Loop | Route appropriate cases to trained peers or qualified professionals. |
| Peer Support | Structured, supervised listener support with escalation rules. |
| Professional Support | Appointment/request workflow for approved providers. |
| Crisis Escalation | Safety-first flow with approved crisis resources and human review. |
| Cybersecurity | Encryption, least privilege, audit logs, monitoring, threat detection and security testing. |
| Admin Dashboard | Aggregate service metrics without exposing private user content by default. |

## 8. SPAG AI Safety & Intelligence Pipeline

**User Input → Validation → Safety Classification → SPAG Analysis → Response Generation → Safety Guardrails → Response → Monitoring**

SPAG is a decision-support and triage layer. It identifies signals that may warrant attention; it does not make irreversible clinical or emergency decisions on its own.

## 9. AI Safety Requirements

- Responses remain within non-clinical emotional-support boundaries.
- The AI must not provide instructions facilitating self-harm or harm to others.
- High-risk signals trigger the approved escalation pathway.
- Safety behavior is tested against adversarial, ambiguous and distressed conversations.
- Human reviewers are available for cases requiring intervention.
- Model performance is evaluated with safety metrics, error analysis and qualified expert review.
- AI changes pass documented safety evaluation before production release.

## 10. Core API Requirements

- POST /auth/register
- POST /auth/login
- POST /auth/logout
- POST /moods
- GET /moods
- POST /journals
- GET /journals
- DELETE /journals/{id}
- POST /ai/conversation
- POST /risk/assess
- POST /support/request
- GET /support/status
- POST /appointments
- GET /appointments
- DELETE /appointments/{id}

## 11. Core Data Entities

**User:** account, consent and access information.

**MoodEntry:** user-created mood check-in.

**JournalEntry:** private journal content.

**Conversation / Message:** AI or human-support interaction data.

**RiskAssessment:** safety/triage output, confidence and review status.

**SupportSession:** peer/professional support interaction and status.

## 12. Security & Privacy Requirements

- TLS for data in transit and strong encryption for sensitive data at rest.
- Role-based access control and least-privilege permissions.
- Secure authentication, session management and optional MFA.
- Strong password hashing such as Argon2id or an approved equivalent.
- Protected audit logging for security-sensitive actions.
- Threat monitoring, rate limiting, WAF/IDS capabilities and incident response.
- Data minimization, retention limits and user access/deletion processes where appropriate.
- Zero Trust principles: continuously verify identity, context and authorization.
- Regular vulnerability scanning, penetration testing and security reviews.

## 13. Technology Architecture

**Mobile:** Flutter/Dart | **Web:** Next.js, React, TypeScript, Tailwind CSS

**Backend:** FastAPI/Python (or NestJS) | **Database:** PostgreSQL | **Cache:** Redis

**AI:** Python, PyTorch/TensorFlow, Hugging Face | **Infrastructure:** Docker, AWS/GCP

**Observability:** Prometheus, Grafana, Sentry/OpenTelemetry | **Realtime:** WebSockets/FCM/APNs

## 14. Institutional Analytics

Dashboards should focus on aggregate/de-identified information such as service demand, response times, support completion and broad wellness trends. Individual private conversations remain protected unless specifically authorized under documented safety/legal policy.

## 15. Success Metrics

| Metric | Purpose |
|---|---|
| Supported Users | Users receiving appropriate support and completing the relevant next step. |
| Escalation Accuracy | Reliability of routing to the appropriate support level. |
| Human Response Time | Time from escalation to human review/response. |
| Support Completion | Percentage completing the recommended support pathway. |
| Safety Incidents | Count and severity of AI/product safety failures. |
| Privacy/Security Incidents | Unauthorized access, exposure or other security events. |

## 16. MVP Acceptance Criteria

- Authentication and authorization are secure and tested.
- Mood tracking and journaling work reliably with appropriate access controls.
- AI responses respect defined non-clinical and safety boundaries.
- Risk/triage behavior is evaluated using representative safety cases and qualified review.
- High-risk cases reach the approved human-review pathway.
- Audit logs and operational monitoring function correctly.
- Security testing is completed before pilot launch.
- Crisis resources and escalation procedures are reviewed by qualified mental-health/safety professionals.
- Privacy notice, consent and data-handling processes are complete before sensitive data collection.

## 17. Development Roadmap

| Stage | Deliverables |
|---|---|
| Phase 0 — Weeks 1–4 | Research, user discovery, safety policy, clinical/safety advisory input, privacy architecture and threat modelling. |
| Phase 1 — Weeks 5–10 | Authentication, mood tracking, journaling, resources, core backend and dashboard. |
| Phase 2 — Weeks 11–16 | AI assistant, SPAG safety pipeline, evaluation harness and guardrails. |
| Phase 3 — Weeks 17–20 | Peer/professional support, escalation and appointment workflows. |
| Phase 4 — Weeks 21–24 | Security hardening, penetration testing, monitoring and pilot readiness. |
| Phase 5 — Months 7–9 | Controlled UNILAG pilot, outcome measurement and iteration. |
| Scale | Other Nigerian universities → West Africa/Africa → international markets, subject to evidence, safety and regulatory readiness. |

## 18. Business Model

Core support should not depend on exploiting vulnerability. Potential revenue includes university SaaS licensing, organizational wellness programs, approved professional-service/platform fees, grants and responsible partnerships. Identifiable mental-health data should not be sold.

## 19. Funding Priorities

- Product and engineering
- Mental-health/safety expertise
- AI safety research and evaluation
- Cybersecurity, privacy and compliance
- Pilot deployment and user research
- Cloud infrastructure and reliability
- Legal, governance and risk management
- Impact measurement
- Operations and responsible growth

## 20. Product Principles

- Safety before growth.

- Privacy before convenience.
- Humans before automation in critical situations.
- Evidence before claims.
- Accessibility before complexity.
- Trust before monetization.
- Prevention before crisis.

## 21. Competitive Advantage

SafeMind's intended differentiation is the integration of prevention, AI-assisted triage, human escalation, privacy-by-design, cybersecurity and institution-level deployment. Sustainable advantage should come from trust, partnerships, localized implementation, validated outcomes and responsible execution—not unsupported clinical claims.

## 22. Product North Star

**Supported Users** — users who receive an appropriate support interaction and successfully complete the relevant next step.

## 23. Final Product Vision

**“We are not building an AI that replaces human care. We are building the missing layer between silence and support.”**

SafeMind is intended to become a scalable, privacy-first mental-wellness infrastructure that helps people reach the right support earlier while keeping qualified humans at the center of critical care.

*Product, clinical, legal, privacy and safety requirements should be reviewed by qualified professionals before real-world deployment.*